1. Startup Landscape Analysis

Before diving into specific analyses, your first task is to understand the overall startup landscape in our database. The executive team needs a snapshot of how many companies have failed (closed down) versus how many are still operating or have been acquired. This will help establish the baseline success rate in the startup ecosystem.

Calculate the number of companies that have been closed down.

```sql
SELECT
    COUNT(*)
FROM
    company
WHERE
    status= 'closed'
```

Result

count
2584

2. Sector Analysis for US Investors

One of our major clients, a US-based VC firm, is considering investments in the media and news space. They've asked us to provide data on how much funding news-related companies from the USA have raised historically, to help them benchmark appropriate investment amounts.

Print the amount of money news-related companies from the USA have raised. Use data from the `company` table. Sort the resulting table by the `funding_total` field in descending order to see the most well-funded companies first.

```sql
SELECT
    funding_total
FROM
    company
WHERE
    category_code= 'news'
    AND country_code= 'USA'
ORDER BY
    funding_total DESC;
```

Result

funding_total
6.22553e+08
2.5e+08
1.605e+08
1.28e+08

3. Analyzing Cash Acquisitions

Our quarterly report includes a section on acquisition trends. The team needs to understand the volume of cash-based acquisitions (as opposed to stock deals) that occurred during the recent post-recession period (2011-2013). This data will help identify whether companies were primarily acquired with cash or other payment methods during this economic recovery period.

Find the total amount of company acquisitions in US dollars. Select only the deals made in cash from 2011 to 2013, inclusive.

```sql
SELECT
    SUM(price_amount)
FROM
    acquisition
WHERE
    term_code= 'cash'
    AND (acquired_at >= '2011-01-01' AND acquired_at <= '2013-12-31');
```

Result

sum
1.37762e+11

4. Identifying Industry Influencers

Our marketing team is preparing an outreach campaign to industry influencers with strong social media presence. They're particularly interested in individuals who brand themselves with "Silver" in their Twitter handles, as this group seems to have significant industry clout. We need to identify these individuals for potential partnerships.

Print the first and last names of people whose Twitter usernames start with 'Silver'. Include their Twitter usernames in the results.

```sql
SELECT
    first_name,
    last_name,
    twitter_username
FROM
    people
WHERE
    twitter_username LIKE 'Silver%'
```

Result

first_name	last_name	twitter_username
Rebecca	Silver	SilverRebecca
Silver	Teede	SilverMatrixx
Mattias	Guilotte	Silverreven

5. Finding Finance Influencers

Following your initial influencer analysis, the marketing team has refined their focus. They're now looking specifically for finance-focused influencers (those with "money" in their Twitter handles) whose last names start with 'K'. This more targeted approach will help them connect with relevant industry voices for our upcoming FinTech investment report.

Print all information about people whose Twitter usernames include the word 'money' somewhere in their name and whose last names start with 'K'.

```sql
SELECT *
FROM
    people
WHERE
    twitter_username LIKE '%money%'
    AND last_name LIKE 'K%';
```

Result

id	first_name	last_name	company_id	twitter_username	created_at	updated_at
63081	Gregory	Kim		gmoney75	2010-07-13 03:46:28	2011-12-12 22:01:34

6. Geographic Investment Analysis

Our global investment clients need to understand funding patterns across different countries. They want to identify which countries attract the most venture capital to help them decide where to focus their international investment strategies. This geographic breakdown will be a key feature in our quarterly global trends report.

For each country, calculate the total amount of money raised by companies registered there. The country code will tell you where each company is registered. Sort the data by sum in descending order to highlight the countries with the most funding.

```sql
SELECT
    country_code,
    SUM(funding_total)
FROM
    company
GROUP BY
    country_code
ORDER BY
    SUM(funding_total) DESC;
```

Result

country_code	sum
USA	3.10588e+11
GBR	1.77056e+10
	1.08559e+10
CHN	1.06897e+10

7. Funding Round Volatility Analysis

Our risk analysis team is examining volatility in funding rounds. They're specifically interested in dates where there was significant variation between the smallest and largest rounds. This indicates days when both very small and very large companies were receiving funding, which could signal unusual market activity. They also want to exclude days where some companies received no funding at all, as that skews the analysis.

Create a table showing the highest and lowest amount of money raised for each date in the `funding_round` table. Include the dates in your results. The resulting table should only have records where the lowest value is not equal to zero or to the highest value.

```sql
SELECT
    funded_at,
    MAX(raised_amount),
    MIN(raised_amount)
FROM
    funding_round
GROUP BY
    funded_at
HAVING
    MIN(raised_amount) <> 0
        AND MIN(raised_amount) <> MAX(raised_amount);
```

Result

funded_at	max	min
2012-08-22	7.5e+07	40000
2010-07-25	9e+06	3.27825e+06
2002-03-01	8.95915e+06	2.84418e+06
2010-10-11	2e+08	28000

Funds that invest in many companies are often seen as having broader networks, while those with fewer investments might have deeper industry expertise. We need to categorize funds by their activity level to help our clients find appropriate partners.

Create a field with three categories:

- `high_activity` — for funds that invest in a hundred or more companies
- `middle_activity` — for funds that invest in between twenty (inclusive) to a hundred companies (exclusive)
- `low_activity` — for funds that invest in fewer than twenty companies

Print all fields from the `fund` table and the new field with categories.

```sql
SELECT
    *,
    CASE
    WHEN invested_companies>=100 THEN 'high_activity'
    WHEN invested_companies BETWEEN 20 AND 99 THEN 'middle_activity'
    WHEN invested_companies <20 THEN 'low_activity'
    END AS activity
FROM
    fund
```

Result

id	name	founded_at	domain
13131			
1	Greylock Partners	1965-01-01	greylock.com
10	Mission Ventures	1996-01-01	missionventures.c

9. Investment Strategy by Fund Activity

Building on our fund activity classification, our research team wants to understand how a fund's investment approach changes based on its activity level. Specifically, we want to know if funds that invest in more companies tend to participate in more funding rounds per company. This will help our clients understand different fund strategies and how broadly or deeply funds typically engage with their portfolio companies.

For each activity category you assigned in the previous task, calculate the average number of funding rounds the fund participated in. Round it to the nearest whole number. Print the categories and the average number of funding rounds. Sort the table by the average in ascending order.

```sql
SELECT
    CASE
    WHEN invested_companies>=100 THEN 'high_activity'
    WHEN invested_companies >=20 THEN 'middle_activity'
    WHEN invested_companies <20 THEN 'low_activity'
    END AS activity,
    ROUND(AVG (investment_rounds)) AS ave_rounds
FROM
    fund
GROUP BY
    activity
ORDER BY
    ave_rounds;
```

Result

activity	ave_rounds
low_activity	2
middle_activity	51
high_activity	252

10. Employee Education Impact on Startup Success

A heated debate has emerged among our clients about whether the educational background of startup employees correlates with company success. Some argue that highly educated teams are more likely to succeed, while others claim education has little impact. To settle this debate with data, we need to compare the education levels of employees at successful companies versus those that closed after limited funding.

We'll start by identifying companies that closed after just one funding round, then analyze the educational backgrounds of their employees.

First, make a list with the names of companies that closed down and had only one funding round while they existed.

Then, find the employees who worked at these companies and join with the education table to analyze their degree types.

Finally, calculate the average number of degrees per employee at these failed startups.

```sql
SELECT AVG(degree_count)
FROM (SELECT
    person_id,
    COUNT(degree_type) as degree_count
FROM
    people
JOIN education ON education.person_id = people.id

WHERE company_id IN (

SELECT id
FROM company
WHERE id IN (SELECT company_id
                                FROM funding_round
                                WHERE is_first_round = 1 AND is_last_round = 1)
                                AND status = 'closed')

GROUP BY person_id) AS avg_degrees;
```

Result

avg
1.23113